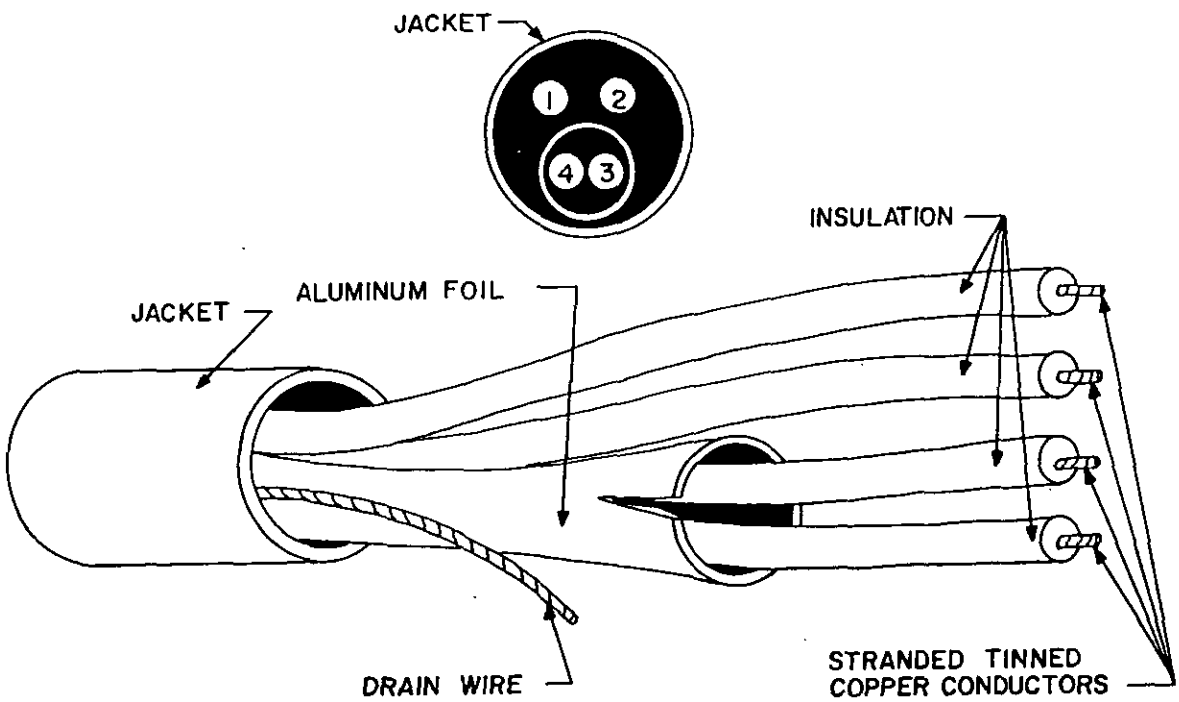
MILITARY SPECIFICATION SHEET

CABLE, SPECIAL PURPOSE, ELECTRICAL, TWO PAIRS OF 20 (7 X 28) AWG, ONE PAIR INDIVIDUALLY SHIELDED

NOTE: NOT FOR AEROSPACE USE.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-49285.



2 PAIR, WITH 1 PAIR INDIVIDUALLY SHIELDED

FIGURE 1. Cable illustration.

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Engineering information: (See table I).

TABLE I. Description, electrical.

Electrical characteristics	Requirements
Nominal capacitance between conductors (shielded pair)	60 pF/ft (picofarads/foot)
Nominal capacitance between the conductor and the shield with one conductor tied to the shield	99 pF/ft
Continuous working voltage	350 V rms maximum between conductors
Shield dc resistance	14 ohms/1,000 feet maximum

TABLE II. Description, physical.

Components	Construction details
Number of pairs	Two pairs - one pair individually shielded
Conductor type and wire size	Stranded tinned copper 20 AWG
Conductor stranding	7 X 28 AWG
Conductor insulation	Polyvinyl chloride
Conductor insulation thickness	0.015 inch nominal thickness
Drain wire type and size	Stranded tinned copper 22 AWG
Drain wire stranding	16 X 34 AWG
Jacket material	Polyvinyl chloride
Jacket thickness	0.028 inch
Finished cable diameter	0.242 inch maximum outside diameter
Cable style (UL)	Not applicable
Tensile strength (jacket)	2,000 pounds per square inch minimum
Elongation (jacket)	150 percent minimum
Overall cabling lay lengths	6.8 twists per foot ±10 percent

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REQUIREMENTS:

Design and construction: (See table II).

Shield color code. The shield color shall be blue.

Shield location and orientation. The polyester aluminum shield is to be located on the outer circumference of the pair with the aluminum foil side outward. Insulation on the interior of the shield is to be complete with no aluminum available for contact.

Drain wire location. Each drain wire is to be spirally located between the shield and jacket, and is to be in continuous contact with the aluminum surface of the foil shield throughout the cable.

Cable temperature rating. The cable temperature rating shall be -20°C to +80°C.

Flammability. The cable shall pass UL 1581 Vertical-Tray flame test requirements.

Conductor color code:

1st conductor: Red
2nd conductor: Black
3rd conductor: Green
4th conductor: White

Cabled on a common axis to reduce the diameter.

Shield integrity test.*

Frequency	Response **
100 kHz	115 dB below reference
500 kHz	104 dB below reference
1 MHz	95 dB below reference
5 MHz	77 dB below reference
10 MHz	67 dB below reference
15 MHz	62 dB below reference
20 MHz	58 dB below reference
25 MHz	54 dB below reference
30 MHz	50 dB below reference

* Any two readings may deviate to a value of 0.9 times the stated limits.

** All values are a minimum numerical value.

Fixture resonance will occur between 40 and 70 MHz.

At 100 MHz: At least 43 dB below reference.

Crosstalk test limits. Crosstalk testing is not required for this configuration.

Durometer hardness. The cable jacket shall have a "Shore A" hardness of 82 ±5.

Part or Identifying Number (PIN): The PIN shall be M49285/19.

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